Learning knowledge from longitudinal data of mammograms to improving breast cancer risk prediction

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ABSTRACT

Breast cancer risk prediction has been attracting more and more attention especially thanks to recent advances in deep learning models. Studies have shown that deep learning on mammograms can provide quantitative imaging features related to breast cancer risk. In breast cancer screening, it is common that a patient has multiple longitudinal mammogram examinations, where the longitudinal imaging data may provide additional information to boost learning of a risk prediction model. In this study, we aim to leverage quantitative imaging features extracted from prior mammogram examinations to augment the training of a risk prediction model through two technical approaches: 1) prior data-enabled transfer learning, and 2) multi-task learning. We evaluated the two approaches on a study cohort of 306 patients in a case-control setting, where each patient has 3 longitudinal screening mammogram examinations. Our results show that both approaches improve risk prediction performance, indicating that additional knowledge can be learned by our approaches from longitudinal imaging data to improve near-term risk prediction.

Key words: Mammograms, breast cancer, risk prediction, longitudinal data, multi-task learning

1. INTRODUCTION

Breast cancer risk prediction has been attracting more and more attention especially thanks to recent advances in deep learning models[1,2]. By using deep learning models, researchers are able to extract imaging features that are more predictive than traditional imaging risk biomarker[3] such as breast density or breast tissue texture. Studies have shown that deep learning on mammograms can provide quantitative imaging features related to breast cancer risk.

Although there are considerable differences between medical image tasks (diagnosis, prediction, etc.) and natural image classification tasks, deep learning models pretrained on massive datasets such as ImageNet are important in medical imaging applications. It has been shown that similarity of tasks can increase the performance on the target task when the source and target tasks are trained on related datasets[4]. In this way, a framework that fuses transfer learning from ImageNet and transfer learning from related tasks can further increase the performance.

When shared knowledge between tasks exists, multi-task learning optimizes the performance by learning all tasks simultaneously. One popular multi-task learning approach in medical applications is to train a classification task and a segmentation task simultaneously using single data point of each patients[5,6]. Also, in recent years multi-task learning

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has been successfully applied to longitudinal clinical data by defining correlated tasks on different timepoints of the data.[7]

In breast cancer screening, it is common that a patient has multiple longitudinal mammogram examinations, where the longitudinal imaging data may provide additional information[8] to boost learning of a risk prediction model. Here, the risk prediction task is defined as using a single prior normal mammogram to predict its outcome (i.e., breast cancer or breast cancer free). In this study, we considered 1-, 2-, and 3-year breast cancer risk prediction using longitudinal mammogram images. It is worth mentioning that we did not combine multiple priors as an input of our models. We expect more recent priors to have more predictive value than older/distant ones (i.e., as the time of diagnosis nears, the mammograms are more likely to contain information related to the cancer development process). This means that risk prediction from older/distant priors is harder than more recent ones. Therefore, we expect that incorporating longitudinal knowledge between a distant prior and a recent prior may enhance the prediction of the recent prior.

Toward this end, our goal is to leverage quantitative imaging features extracted from prior mammogram examinations to augment the training of a risk prediction model through two technical approaches: 1) prior data-enabled transfer learning, and 2) multi-task learning. To compare whether such a knowledge can be used to boost the training of the model, we performed single independent learning as well to serve as a baseline, which only used ImageNet pretrained model as a weight initializer.

2. MATERIALS AND METHODS

2.1 Study cohort and dataset

In this study, with IRB approval we retrospectively collected contralateral digital screening mammograms in 306 patients who underwent general population screening for breast cancer. This cohort includes 153 women with biopsy-proven unilateral invasive breast cancer and 153 controls. The controls had negative findings and remained cancer-free for at least one-year after the most recent examination. The controls were age-matched to the cancer cases with a ±1 year range. All the 306 women had 3 screening mammogram examinations (within 3 to 6 years) before the final outcome (cancer or control). We used prior-1 to prior-3 to denote the most recent and most distant prior screening mammograms. Our data only included the cranio-caudal (CC) views for each patient. Given a case-control pair, we included the cancer-affected breast for the case and we selected the image of the same breast side from the control. **Figure 1** shows an example of 3 normal priors along with the current image. It's worth mentioning that in this study we only used these prior examinations to evaluate the risk of developing cancer.

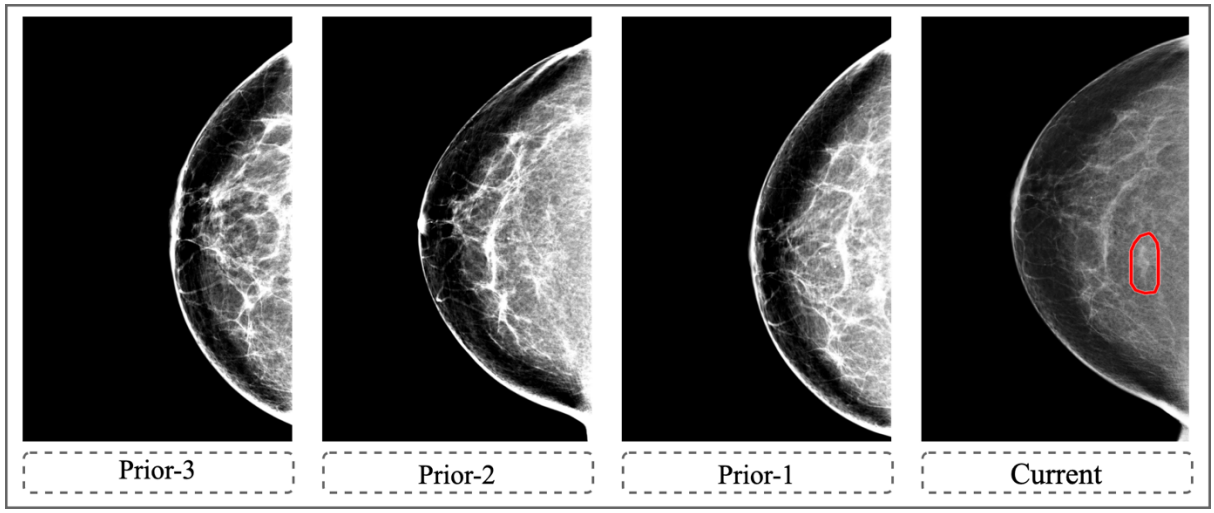


Figure 1. Example of 3 timesteps of longitudinal mammogram examinations belonging to a 54-year-old woman diagnosed with Ductal Carcinoma In-Situ. Prior-1 through prior-3 show the most recent to the most distant prior screening mammogram. We show the current image only to illustrate the appearance of the malignant tissue after 3 normal prior mammograms.

2.2 Methodology

In this work, we investigated three approaches to learn prior knowledge from longitudinal examinations. **Figure 2** shows the whole study design scheme with three tasks. The first approach serves as a baseline: we simply train three separate deep learning models to perform each of the three risk prediction tasks. The second approach initially trains a model on task-3 (risk prediction from the oldest prior), then fine-tunes the obtained model on task-2, and finally fine-tunes this model on task-1. Since this approach uses transfer learning to propagate information from one prior to the next, we call it prior data-enabled transfer learning. Our third approach uses multi-task learning to jointly learn all three tasks.

As a first step, we randomly split the patients into three groups: train, validation, and test. We used 183 (60%) patients for training, 61 (20%) patients for validation, and 62 (20%) patients for testing. Since we split the data at patient level, the training/validation/test groups are the same for each task.

To preprocess the images before further analysis, we used the publicly available LIBRA[9] package to automatically segment the breast region. We resized images to 512x512 pixels. For deep learning model, we used InceptionResNet-V2[10] as our backbone network. We implemented our experiments using Tensorflow 1.15 and Keras 2.0.4. We ran all experiments on NVIDIA V100 GPUs from the Pittsburgh Supercomputing Center (PSC).

We adopted Adam as an optimizer with learning rate 0.0001 and trained the models for 200 epochs. For the loss function, we used binary cross-entropy. To increase the variability of the input data we used vertical and horizontal flip data augmentation in the training process.

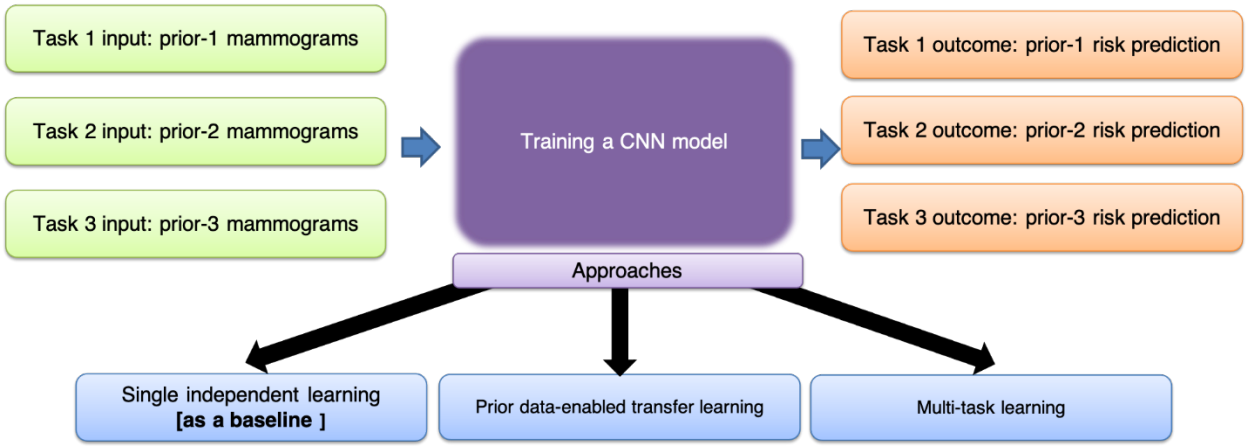


Figure 2. Our study design scheme showing the three proposed approaches for risk prediction. The first approach serves as a baseline as we trained each task individually. In the other two approaches, features from different priors are leveraged by either transfer learning (features learned from one prior are fine-tuned from those learned from another prior) or multi-task learning.

2.3 Prior Data-Enabled Transfer Learning

With this approach, we started by training a convolutional neural network (CNN) model on task-3. We then fine-tuned the obtained model on task-2, and further fine-tune this second model on task-1. In this way, the information extracted from prior-3 is propagated to prior-2 and eventually to prior-1. **Figure 3** shows a representation of this approach.

2.4 Multi-Task Learning

In this approach, we used multi-task learning to predict breast cancer risk. We incorporated three individual models with shared weights on all layers except for the last dense layer which was responsible for binary classification for that specific task. **Figure 4** shows a graphic representation of this approach.

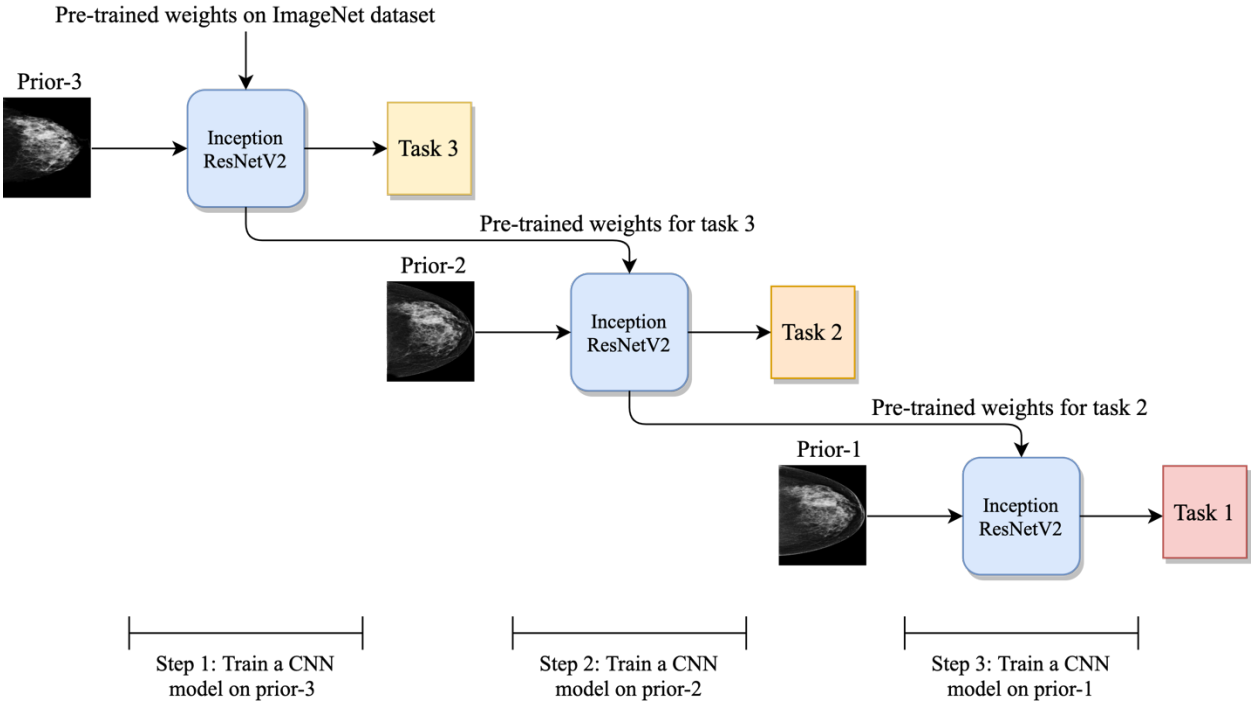


Figure 3. Depiction of our proposed data-enabled transfer learning approach for breast cancer risk prediction. Task 1/2/3 was to predict 1/2/3-year risk of developing breast cancer. In step 1 of this scenario, we used ImageNet model as an initializer to train task-3. Then in step 2, we picked the model for task-3 and used it as an initializer for training task-2. Following the same process, the best model for task-2 was chosen as a weight initializer for task-1. The training, test and validation sets were the same through all the steps.

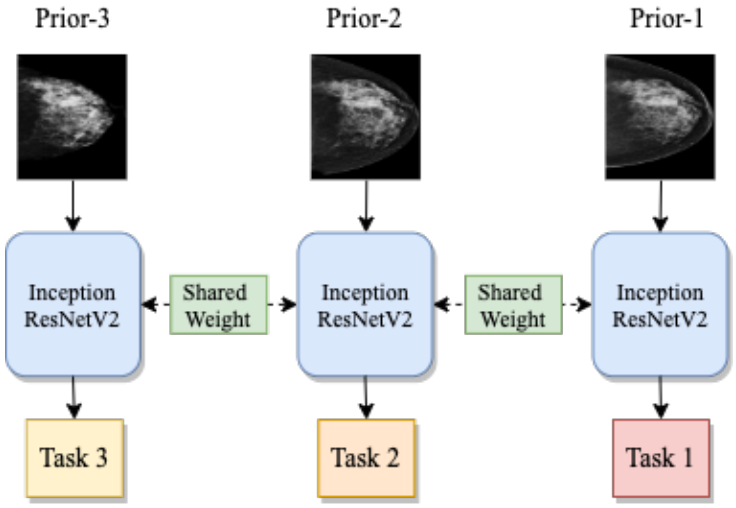


Figure 4. An overview of our proposed multi-task learning model for breast cancer risk prediction. Task 1/2/3 was to predict 1/2/3-year risk of developing breast cancer.

3. RESULTS

In this section, we reported all the experiment results, including: 1) single independent tasks for each prior using ImageNet pretrained model (**Table 1)**, which serves as a baseline as it does not include any knowledge transferring; 2) single tasks using prior data-enabled transfer-learning (**Table 2);** and 3) multi-task learning (**Table 3)**. We reported model performance based on area under the ROC curve (AUC) metric.

In **Table 2** we followed the same training as in **Table 1** except using different data for pretraining. Specifically, for (d) we used the best model on prior-2 risk prediction task as a pretrained network, then fine-tuned it on prior-1 risk prediction model. For (e), we used the best model on prior-3 prediction task as a pretrained network, then fine-tuned it on prior-2 prediction model. Finally, for (f) we used the best model from (e) to first incorporate the knowledge from prior-3 to prior-2 and then on prior-1 risk prediction task.

As we can see from the three tables, higher AUC values were achieved in both of the two approaches (**Tables 2 and 3**) than the baseline method (**Table 1**). Further, the prior data-enabled transfer learning (**Table 2**) outperformed the multi-task learning (**Table 3**).

Table 1. Results of single independent tasks using ImageNet as a pretrained model.

Task (baseline)	AUC
(a) Prior-1 risk prediction	0.60
(b) Prior-2 risk prediction	0.55
(c) Prior-3 risk prediction	0.50

Table 2. Results of single tasks using prior data-enabled transfer learning.

Task	AUC
(d) Prior-1 risk prediction (transfer learning from Prior-2)	0.65
(e) Prior-2 risk prediction (transfer learning from Prior-3)	0.61
(f) Prior-1 risk prediction (transfer learning from Prior-3)	0.62
(g) Prior-1 risk prediction (transfer learning from (e))	**0.66**

Table 3. Multi-task learning results.

Task	AUC
(h) Prior-1 risk prediction	0.60
(i) Prior-2 risk prediction	0.60
(j) Prior-3 risk prediction	0.62

4. DISCUSSION AND CONCLUSION

In this work, we developed a prior data-enabled transfer learning as well as a multi-task learning framework to investigate whether potential knowledge in prior longitudinal mammogram images can be learned to benefit the prediction of near-term breast cancer risk. Based on our dataset we defined 3 tasks including prior-1, prior-2 and prior-3 risk prediction. We conducted experiments using 306 patients in a case-control setting.

As we hypothesized, **Table 1** shows how risk prediction from prior-3 is harder than risk prediction from prior-2 (AUC: 0.50 v.0.55), and risk prediction from prior-2 is, in turn, harder than risk prediction from prior-1 (AUC: 0.55 v. 0.60).

As expected, our results showed encouraging improvement in the AUC values in the two approaches we implemented compared to the baseline results where no knowledge is learned from the prior examinations. Our study represented a

new approach to leverage longitudinal imaging data and distill potential spatiotemporal knowledge conveyed in these data to improve deep learning models for breast cancer risk prediction.

Results also showed when two tasks were closer (e.g., [prior-3 and prior-2], [prior-2 and prior-1]), they shared more knowledge. To be more specific, using transfer learning from task-3 to task-2, performance on task-2 increases from 0.55 to 0.6 ((b) and (e) in **Tables 1 and 2**). In addition, using transfer learning from task-2 to task-1 improved results on task-1 from 0.6 to 0.65 ((a) and (d) in **Tables 1 and 2**). In both cases, we observed higher improvement in AUC than with transfer learning from prior-3 on prior-1 only (from 0.6 to 0.62, (a) and (f) in **Tables 1 and 2**). Furthermore, involving prior-3 and prior-2 in the pretraining process can convey more knowledge to task-1 (g) rather than using them separately for pre-training ((d) and (f)).

In conclusion, we showed the concept of incorporating the knowledge from prior mammograms in longitudinal study and we demonstrated spatiotemporal information extracted by our framework can boost the overall performance in risk prediction using sequential mammograms. We are working on developing new models that combine multiple prior mammogram examinations for breast cancer risk prediction.

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